KAIZEN MATHS

Lesson starter

Equation of a Circle

Equation Forms

Name _____

S1

Write the equation of the circle with centre (1, 1) and radius 5.

S2

A circle has diameter endpoints (-2, 0) and (4, 0). Write its equation.

S3

Write the equation of the circle with centre (-3, 6) and radius 5.

S4

A circle has diameter endpoints (2, -6) and (6, -2). Write its equation.

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GEO-9D18-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Equation Forms

Name

E1

Write the equation of the circle with centre (4, -5) and radius 2.

E2

A circle has diameter endpoints (-1, -5) and (9, 3). Write its equation.

E3

Write the equation of the circle with centre (-6, -3) and radius 9.

KAIZEN MATHS

Exit ticket

Equation of a Circle

Equation Forms

Name

E1

Write the equation of the circle with centre (4, -5) and radius 2.

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A circle has diameter endpoints (-1, -5) and (9, 3). Write its equation.

E3

Write the equation of the circle with centre (-6, -3) and radius 9.

GEO-9D18-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Equation Forms

Name

E1

Write the equation of the circle with centre (4, -5) and radius 2.

E2

A circle has diameter endpoints (-1, -5) and (9, 3). Write its equation.

E3

Write the equation of the circle with centre (-6, -3) and radius 9.

GEO-9D18-L2 · E

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Equation of a Circle

Equation Forms

Name

E1

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GEO-9D18-L2 · E

Show your working

Teacher answers & working

Equation of a Circle · Equation Forms

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S1

Write the equation of the circle with centre (1, 1) and radius 5.

Answer:

$(x - 1)^2 + (y - 1)^2 = 25$

Working:

Use $(x - a)^2 + (y - b)^2 = r^2$.

Here a = 1, b = 1, and r = 5.

Substitute the centre and radius.

$(x - 1)^2 + (y - 1)^2 = 25$

S2

A circle has diameter endpoints (-2, 0) and (4, 0). Write its equation.

Answer:

$(x - 1)^2 + y^2 = 9$

Working:

The centre is the midpoint of the diameter.

Centre = (1, 0) = (1, 0).

Find r^2 using one endpoint and the centre.

$r^2 = (4 - 1)^2 + (0 - 0)^2 = 9$.

Substitute into $(x - a)^2 + (y - b)^2 = r^2$.

$(x - 1)^2 + y^2 = 9$

S3

Write the equation of the circle with centre (-3, 6) and radius 5.

Answer:

$(x + 3)^2 + (y - 6)^2 = 25$

Working:

Use $(x - a)^2 + (y - b)^2 = r^2$.

Here a = -3, b = 6, and r = 5.

Substitute the centre and radius.

$(x + 3)^2 + (y - 6)^2 = 25$

Teacher answers & working

Equation of a Circle · Equation Forms

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S4

A circle has diameter endpoints (2, -6) and (6, -2). Write its equation.

Answer:

$(x - 4)^2 + (y + 4)^2 = 8$

Working:

The centre is the midpoint of the diameter.

Centre = (4, -4) = (4, -4).

Find r^2 using one endpoint and the centre.

$r^2 = (6 - 4)^2 + (-2 - -4)^2 = 8$.

Substitute into $(x - a)^2 + (y - b)^2 = r^2$.

$(x - 4)^2 + (y + 4)^2 = 8$

E1

Write the equation of the circle with centre (4, -5) and radius 2.

Answer:

$(x - 4)^2 + (y + 5)^2 = 4$

Working:

Use $(x - a)^2 + (y - b)^2 = r^2$.

Here a = 4, b = -5, and r = 2.

Substitute the centre and radius.

$(x - 4)^2 + (y + 5)^2 = 4$

E2

A circle has diameter endpoints (-1, -5) and (9, 3). Write its equation.

Answer:

$(x - 4)^2 + (y + 1)^2 = 41$

Working:

The centre is the midpoint of the diameter.

Centre = (4, -1) = (4, -1).

Find r^2 using one endpoint and the centre.

$r^2 = (9 - 4)^2 + (3 - -1)^2 = 41$.

Substitute into $(x - a)^2 + (y - b)^2 = r^2$.

$(x - 4)^2 + (y + 1)^2 = 41$

Teacher answers & working

Equation of a Circle · Equation Forms

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E3

Write the equation of the circle with centre (-6, -3) and radius 9.

Answer:

$(x + 6)^2 + (y + 3)^2 = 81$

Working:

Use $(x - a)^2 + (y - b)^2 = r^2$.

Here a = -6, b = -3, and r = 9.

Substitute the centre and radius.

$(x + 6)^2 + (y + 3)^2 = 81$